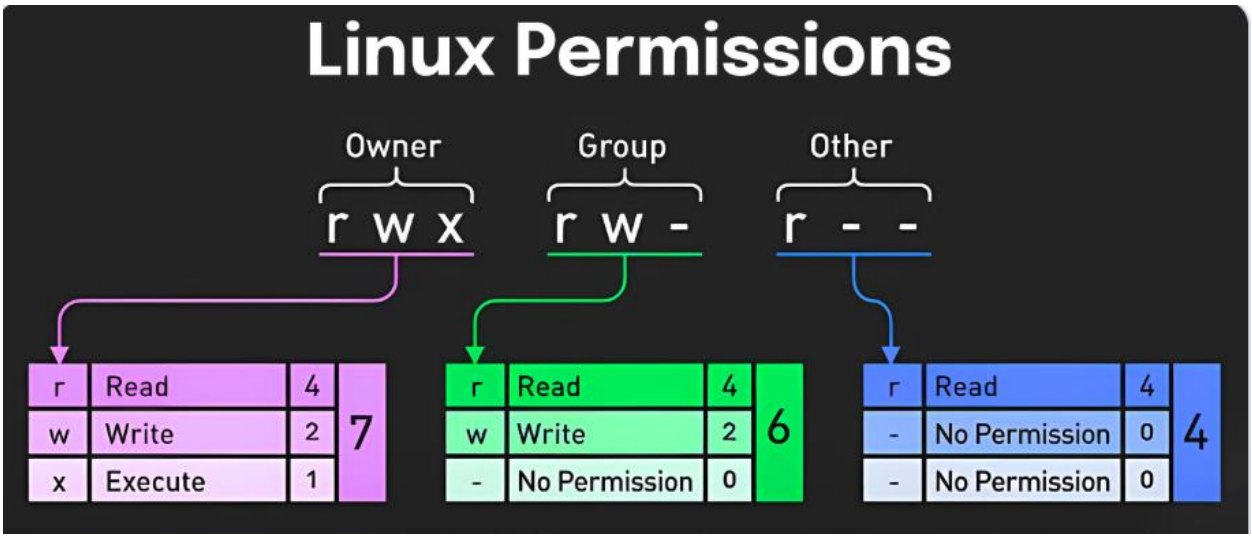
Linux Permissions

Linux file permissions form the foundation of the system's security model. They define who can read, write, or execute files and directories, ensuring only authorized users or processes can access sensitive data.

1. The Three Basic Permissions

Every file or directory has three types of permissions:



2. Ownership and Permission Groups

Permissions are assigned to three categories of users:

- **User (Owner):** The person who created the file.
- **Group:** Users belonging to a shared group (e.g., "developers" or "admins").
- **Others:** Everyone else on the system.

File Permission: Operation Chart

Operators	Definition
`+`	Add permissions
`-`	Remove permissions
`=`	Set the permissions to the specified values

Note: *All these permissions are being granted at three different levels based on their group.*

What are Permission Groups in Linux

First, you must think of those nine characters as three sets of three characters (see the box at the bottom). Each of the three "rwx" characters refers to a different operation you can perform on the file.

1. **Owners:** These permissions apply exclusively to the individuals who own the files or directories.
2. **Groups:** Permissions can be assigned to a specific group of users, impacting only those within that particular group.
3. **All Users:** These permissions apply universally to all users on the system, presenting the highest security risk. Assigning permissions to all users should be done cautiously to prevent potential security vulnerabilities.

```
---      ---      ---
rwx      rwx      rwx
user     group    other
```

User, Group, and others Option in Linux File Permission

Reference	Class	Description
`u`	**user**	The user permissions apply only to the owner of the file or directory, they will not impact the actions of other users.
`g`	**group**	The group permissions apply only to the group that has been assigned to the file or directory, they will not affect the actions of other users.
`o`	**others**	The other permissions apply to all other users on the system, this is the permission group that you want to watch the most.
`a`	**All three**	All three (owner, groups, others)

How to Check the Permission of Files in Linux

Let's dive in to understand the possible methods to check all the desired details of a file including "File Permission"

The 'ls' Command

Here's the command to execute it within the terminal. Let's show you with an example:

Input:

We're taking 'NarX' as a default file name:

```
ls -l NarX.txt
```

Output:

```
-rw-r--r-- 1 user group 46 Apr 14 16:37 NarX.txt
```

The above command represents these following information:

1. The first character = **'-'**, which means it's a file **'d'**, which means it's a directory.
2. The next nine characters = (rw-r--r--) show the security
3. The next column shows the owner of the file.
4. The next column shows the group owner of the file. (which has special access to these files)
5. The next column shows the size of the file in bytes.
6. The next column shows the date and time the file was last modified.

The 'namei' Command

The 'namei' command is used to check the file path through layer of folder's path. Here's the command to execute it within Terminal:

Here, we've taken 'path' as " root@anonymous-VirtualBox:~# " and file name as "hoops"

```
namei -l /path/to/your/file
```

The 'stat' Command

Unlike 'ls -l' command, the "stat" command is used to pin point the file location. Here's how you can do it:

We're taking file name as "hoops"

```
stat hoops
```

Output:

```
 File: example.txt
  Size: 2210       Blocks: 8          IO Block: 4096  regular file
Device: 802h/2050d    Inode: 1288496     Links: 1
Access: 2024-11-18 10:50:56.000000000 +0000
Modify: 2024-11-18 10:50:56.000000000 +0000
Change: 2024-11-18 10:50:56.000000000 +0000
 Birth: -
```

How to Change Permissions in Linux

The command you use to change the security permissions on files is called "**chmod**", which stands for "change mode" because the nine security characters are collectively called the security "mode" of the file. You can modify permissions using **symbolic notation** or **octal notation**.

Symbolic Notation

Symbolic notation allows you to **add, remove,** or **set permissions** for specific users. Let's understand this using different example below:

Example 1: To Change File Permission in Linux

If you want to give "execute" permission to the world ("other") for file "xyz.txt", you will start by typing.

```
chmod o
```

Now you would type a '+' to say that you are "adding" permission.

```
chmod o+
```

Then you would type an 'x' to say that you are adding "execute" permission.

```
chmod o+x
```

Finally, specify which file you are changing.

```
chmod o+x xyz.txt
```

You can see the change in the picture below.

Example :

The code adds read(r) and write(w) permission to both user(u) and group(g) and revoke execute(x) permission from others(o) for the file abc.mp4.

```
chmod ug+rw,o-x abc.mp4
```

Octal Notations Permissions in Linux

The octal notation is used to represent file permission in Linux by using three user group by denoting 3 digits i.e.

- user
- group
- other users

Here's how to permissions are mapped:

- **Read (r) = 4**
- **Write (w) = 2**
- **Execute (x) = 1**

Permissions for owner, group, and others are represented by a three-digit octal value. The sum of permissions for each group gives the corresponding number.

Reference:

`chmod o`

Now you would type a '+' to say that you are "adding" permission.

`chmod o+`

Then you would type an 'x' to say that you are adding "execute" permission.

`chmod o+x`

Finally, specify which file you are changing.

`chmod o+x xyz.txt`

You can see the change in the picture below.

Example:

The code adds read(r) and write(w) permission to both user(u) and group(g) and revoke execute(x) permission from others(o) for the file abc.mp4.

```
chmod ug+rw,o-x abc.mp4
```

You can also use octal notations like this.

octal notations

Using the octal notations table instead of 'r', 'w', and 'x'. Each digit octal notation can be used for either of the group 'u', 'g', or'o'.

Security Permissions in Linux

The combination for the permissions are r,w,x, and -. Let's understand this briefly:

For example: "rw- r-x r--"

- **"rw-"**: the first three characters `rw-`. This means that the owner of the file can "read" it (look at its contents) and "write" it (modify its contents). We cannot execute it because it is not a program but a text file.
- **"r-x"**: the second set of three characters "r-x". This means that the members of the group can only read and execute the files.
- **"r--"**: The final three characters "r--" show the permissions allowed to other users who have a UserID on this Linux system. This means anyone in our Linux world can read but cannot modify or execute the files' contents.

Special Permissions in Linux

Besides usual methods, Linux also offers special permission types to have more complex control over files.

1. The 'setuid' Command

The SET User ID permission allows user to execute programs with the privileges of its owner. Below is the example for the same:

```
chmod u+s program
```

2. The 'setgid' Command

The Set Group ID permission allows files to run under file's group permissions (or ensures the files created in a directory inherits the group of the directory). Here's the command for the same:

```
chmod g+s directoryname
```

3. The 'sticky bit' Command

This allows the user (only owner) to delete or rename files within the directory (regardless of other user's permissions). Here's a command for the same:

```
chmod +t directoryname
```

How to Set File Permissions for a Specific User

To set permissions for a specific user or group:

By using chown

Use chown to change file ownsership:

```
chown user:group file.txt
```

By using chmod

Use chmod to modify permissions:

```
chmod 755 file.txt
```

realword use case :

- **755 → executable (scripts, folders)**
- **644 → normal files (read/write only for owner)**